

HyperGNNs for Multi-Modal Classification and Severity Analysis of Neurodegenerative Disorders

K Bhavishh Raju ^{1*†}, K Musadiq Pasha ^{2†},
Mohammed Saqlain ^{3†}, Nishaan Padanthaya ^{4†},
Dr. Jayashree R ^{5†}

^{1*}Dept. of Computer Science Engineering (AI & ML), PES University,
Bangalore, India.

*Corresponding author(s). E-mail(s): bhavishk17@gmail.com;
saqlain.contact@gmail.com;

Contributing authors: kmusadiqpasha@gmail.com;
nishaanpj@gmail.com; jayashree@pes.edu;

[†]These authors contributed equally to this work.

Abstract

Neuro-degenerative disorders, particularly Alzheimer’s Disease (AD), pose a significant challenge in early diagnosis and severity assessment due to overlapping symptoms with conditions such as Mild Cognitive Impairment (MCI) and Cognitively Normal (CN) conditions. Accurate differentiation between these stages is essential for timely intervention but remains difficult due to the progressive and heterogeneous nature of these disorders. Traditional machine learning models struggle to effectively integrate diverse data modalities, such as medical imaging (MRI) and clinical tabular data. This study proposes Hypergraph Neural Networks (HyperGNNs) based framework to enhance multi-modal classification and disease severity modeling. By representing complex patient relationships as hypergraphs, our approach aims to improve diagnostic accuracy, reduce misdiagnosis, and provide an interpretable framework for understanding disease progression (CN to MCI to AD). To ensure clinical interpretability and transparency, we incorporate explainability techniques such as SHAP for tabular values and Grad-CAM for MRI features to ensure model transparency, enabling clinicians to understand key features influencing predictions. The model will be evaluated on standard neuro-imaging datasets and clinical records, offering potential applications in personalized medicine and early intervention strategies.

Keywords: Hypergraph Neural Networks, Alzheimer’s Disease, Multi-modal Learning, Explainable AI, Medical Image Analysis, Disease Severity Modeling

1 Introduction

Neurodegenerative disorders represent a class of chronic and progressive conditions characterized by the gradual deterioration of neuronal structure and function. Among them, Alzheimer’s Disease (AD) is the most prevalent, affecting millions worldwide and posing a significant burden on healthcare systems and affected individuals. Early diagnosis of AD and related conditions such as Mild Cognitive Impairment (MCI) is critical for timely intervention and improving patient outcomes. The early-stage identification and severity classification of AD remains inherently challenging due to the overlapping symptoms and the heterogeneous nature of disease progression across individuals. Current diagnostic rely heavily on neuro-imaging techniques, cognitive assessments, and clinical history and are often limited by delayed manifestation of symptoms, and fragmented data sources.

In recent years, Artificial Intelligence and Machine Learning have shown immense promise in assisting with early and accurate diagnosis, but often face several critical limitations. Most models operate on single data modalities (either imaging or clinical data) or employ simple concatenation strategies that fail to capture complex interactions between different data types. The synergistic information from combining structural brain changes with demographic and clinical variables remains largely underutilized. Traditional supervised learning treats each patient as an independent sample, ignoring potentially valuable relational information. Patients sharing similar demographic profiles, genetic risk factors, or disease trajectories may exhibit common patterns that could improve generalization and diagnostic accuracy. Deep learning models often function as “black boxes,” providing predictions without transparent reasoning. In the medical domain, where diagnostic decisions have serious consequences, understanding which features drive predictions is essential for building clinician trust and ensuring appropriate model deployment. Although classification has been the main focus, continuous severity assessment and modeling of disease progression pathways ($CN \rightarrow MCI \rightarrow AD$) remain underexplored which could help in patient monitoring and intervention planning.

To address these limitations, we propose a Hypergraph Neural Network (HyperGNN) framework that advances multi-modal learning for neuro-degenerative disease classification and severity analysis. We model patient groups as hypergraphs where nodes represent individual patients and hyperedges connect groups of patients sharing similar characteristics. The data we used is THE Alzheimer’s Disease Neuroimaging Initiative (ADNI) which consists of the structural MRI scans fused with the patient demographic and clinical tabular data. We integrate SHAP for feature importance analysis of tabular variables and Grad-CAM for highlighting discriminative brain regions in MRI scans. The combination of multi-modal fusion, hypergraph representation, and explainability is expected to provide a more accurate, interpretable, and robust system for aiding clinical decision-making.

2 Literature Survey

The diagnosis and classification of Alzheimer’s Disease and related neurodegenerative disorders have attracted substantial research attention in recent years, with approaches involving deep learning, graph neural networks, multi-modal fusion, and explainable AI.

Early applications of deep learning to AD diagnosis primarily focused on single-modality approaches using Convolutional Neural Networks (CNNs) for structural MRI analysis. El-Assy et al. (El-Assy et al. 2024) proposed a custom 11-layer CNN architecture optimized for classifying MRI data into CN, MCI, and AD categories. Their approach employs two parallel CNN streams with distinct filter sizes and pooling layers, concatenated at the classification stage. Despite its architectural simplicity, the model achieved approximately 95% accuracy across 3-way, 4-way, and 5-way classification tasks on the ADNI dataset. However, single-modality approaches inherently limit diagnostic potential by excluding complementary information available from other imaging modalities and clinical data sources. This limitation has driven recent research toward multi-modal integration strategies.

Adarsh et al. (Adarsh et al. 2024) introduced a Multi-Kernel Semi-Classical Dynamic Learning (MKSCDDL) kernel integrated with CNN architectures for heterogeneous data fusion. Their framework combines MRI and PET modalities while incorporating Scandent Decision Trees to handle inter-subject variability and improve explainability. The model achieved 98.27% classification accuracy for AD, MCI, and CN categories, emphasizing both diagnostic performance and transparent decision-making.

Ma et al. (Ma et al. 2020) addressed differential diagnosis among Alzheimer’s Disease, Frontotemporal Dementia (FTD), and normal aging using Generative Adversarial Networks (GANs) for data augmentation combined with multi-scale, multi-type feature extraction. Their architecture consists of six independent multilayer perceptron blocks trained on different feature scales and types, with outputs combined for three-class classification. The GAN-augmented model achieved 88.28% accuracy, improving upon the 83.06% baseline SVM performance. This work highlights the benefits of data augmentation and multi-scale representation learning, particularly for imbalanced medical datasets.

Tang et al. (Tang et al. 2024) proposed an improved Transformer-based architecture combining MRI and PET scans for AD diagnosis. Their pipeline employs 3D CNNs for deep feature extraction from each modality, followed by a modified Transformer module that progressively learns global correlations among extracted features. This work demonstrates the effectiveness of attention mechanisms in capturing cross-modal dependencies, though the computational complexity of Transformer architectures may limit scalability.

Lao et al. (Lao et al. 2024) proposed a structural GCN framework for AD diagnosis using graph representations of brain regions of interest (ROIs). Their approach applies 3D Discrete Wavelet Transform (3D-DWT) to extract features from 90 brain ROIs across sMRI and FDG-PET modalities. A graph is constructed with nodes representing ROIs, edges encoding spatial relationships, and node features derived from 3D-DWT. The GCN-based model achieved 93.12% accuracy showing the importance

of using structural graphs. Subaramya et al. (Subaramya et al. 2022) extended this concept using diffusion MRI (dMRI) and T1-weighted MRI to construct structural brain networks. Their framework segments the brain into 80 regions and obtained 97% accuracy in distinguishing AD from CN subjects.

Sang and Li (Sang and Li 2024) utilized Diffusion Tensor Imaging (DTI) to construct brain networks where edges reflect white matter connectivity. Their approach integrates GCN with a self-attention mechanism to weight critical connections adaptively. The model achieved 87.5% accuracy.

Medical datasets frequently exhibit class imbalance, with minority classes under-represented. Ghorbani et al. (Ghorbani et al. 2022) addressed this challenge through Re-Weighted Adversarial Graph Convolutional Networks (RA-GCN), which dynamically assign sample weights to reduce bias. Their work involved integrating an attention mechanism to weight node neighborhoods adaptively, prioritizing more relevant nodes during message passing. RA-GCN outperformed baseline GCN models, particularly in minority class recall.

Most existing studies focus on cross-sectional analysis, limiting their ability to model disease progression over time. Kim et al. (Kim et al. 2021) proposed an interpretable temporal GNN integrating graph convolutional networks with recurrent neural networks (RNN) for prognostic prediction using longitudinal neuroimaging data. Their model analyzes brain connectivity changes derived from diffusion MRI across multiple time points to predict disease progression. The temporal GNN achieved 53.5% accuracy in predicting AD progression at 24 months, outperforming deep neural networks and SVM. Additionally, using GNNExplainer, the authors identified key brain regions (superior frontal, insula) contributing to predictions.

Pan et al. (Pan et al. 2021) introduced Hypergraph Generative Adversarial Networks (HGGAN) for characterizing multi-modal brain connectivity using resting-state fMRI and DTI. The approach achieved 93.02% accuracy across six classification tasks when combined with GCN, identifying seven key brain regions affected by AD, providing superior characterization of multi-modal brain connectivity. Hao et al. (Hao et al. 2024) proposed Weighted Hypergraph Convolution Networks (WHGCN) for longitudinal sMRI analysis. Their framework constructs patient hypergraphs at each time point, where hyperedges connect groups of patients with similar neuroimaging and clinical features. Both these works showed better performance in identifying key characteristics of the brain being affected by AD highlighting the effectiveness of hypergraph structures.

The "black-box" nature of deep learning models poses significant challenges for clinical deployment, where transparency and interpretability are essential for building clinician trust and ensuring accountability. Some of the work have tried to incorporate different kinds of explainability to their approaches. Adarsh et al. (Adarsh et al. 2024) utilized Scandent Decision Trees to provide interpretable insights into neuroanatomical regions linked to cognitive impairment. Tang et al. (Tang et al. 2024) employed model-based visualization to localize AD-related brain regions, while Sang and Li (Sang and Li 2024) demonstrated that self-attention mechanisms in GCN-based frameworks can identify critical white matter connectivity patterns. Kim et al. (Kim et al. 2021) used GNNExplainer to identify brain regions contributing to AD progression predictions.

Table 1 summarizes the key characteristics of related approaches.

Table 1: Comparison of Related Work on AD Classification

Study	Method	Modality	Classes	Accuracy	XAI
Adarsh et al. (Adarsh et al. 2024)	MKSCDDL-CNN	MRI+PET	3	98.27%	Yes
Kim et al. (Kim et al. 2021)	Temporal GNN	dMRI	2	53.50%	Yes
Our Proposed Methodology	HyperGNN	sMRI+Tabular	6	94.00%	Yes

3 Methodology

The proposed framework for neuro-degenerative disease classification and severity assessment is designed as a multi-stage pipeline. It begins with pre-processing and feature extraction from raw structural MRI (sMRI) scans. These engineered features are then mapped and merged with tabular data containing the demographics of the patients. This complete merged and processed data is used to construct a feature-centric hypergraph for the primary classification task using a Hypergraph Neural Network (HyperGNN). In parallel, an unsupervised clustering approach is employed to model disease severity, which is then used to train a separate predictive model. Finally, explainability techniques are applied to both the classification and severity models to ensure clinical interpretability. The overall framework is illustrated in Figure 1.

3.1 Data Preprocessing

We utilize the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset, which provides multi-modal data including structural MRI (sMRI) scans in DICOM format, alongside corresponding demographic and clinical tabular data for each patient.

The dataset comprises three subject groups based on cognitive status:

1. **Cognitively Normal (CN)** : No signs of clinically significant memory problems.
2. **Subjective Memory Concern (SMC)** : This stage is characterized by a person’s subjective feeling that their memory or other cognitive abilities are declining, even though objective neuropsychological tests may be normal.
3. **Mild Cognitive Impairment (MCI)** : An intermediate stage between the expected cognitive declines of normal aging and the more serious decline of dementia.
4. **Early Mild Cognitive Impairment (EMCI)** : A less severe form of MCI, characterized by milder cognitive deficits.
5. **Late Mild Cognitive Impairment (LMCI)** : A more severe form of MCI, with cognitive impairment closer to the dementia threshold.
6. **Alzheimer’s Disease (AD)** : A progressive, irreversible decline in cognitive abilities.

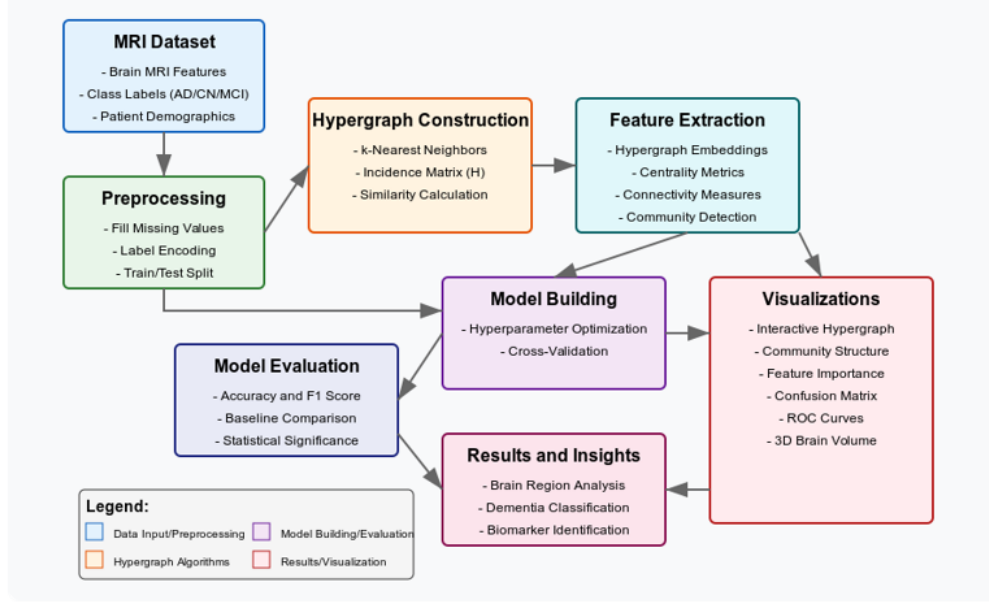


Fig. 1: Overview of the proposed HyperGNN framework for neurodegenerative disease classification and severity analysis.

The pre-processing process of the data acquired begins by reading the 2D DICOM slices for each patient scan and stacking them to reconstruct a 3D volumetric representation of the brain. Once the Image reconstruction is done, we move to Brain Extraction. A crucial step to ensure that subsequent analysis focuses solely on relevant brain tissue is Brain Extraction, which isolates the brain from surrounding non-brain tissues like the skull, scalp, and eyes. This is achieved through a semi-automated pipeline:

3.1.1 Intensity Normalization

The 3D volume's voxel intensities are first normalized to a standard range of $[0, 1]$ to ensure consistency across different scans.

3.1.2 Thresholding and Mask Generation

A global intensity threshold is applied to create an initial binary brain mask, separating the brighter brain tissue from the darker background.

3.1.3 Morphological Refinement

To enhance the mask's quality, a series of morphological operations are performed. Binary closing is used to fill small, spurious holes within the brain region, and binary

opening is applied to remove small, disconnected noise components (artifacts) outside the main brain mass.

3.1.4 Largest Connected Component Analysis

Finally, to ensure the mask represents only the brain, we identify the largest 3D connected component within the binary mask. All smaller, disconnected components are discarded, resulting in a clean and accurate brain mask.

After obtaining the brain mask, we next perform brain tissue segmentation. With the brain region isolated, we segment the intracranial volume into its three primary tissue types: Cerebrospinal Fluid (CSF), Gray Matter (GM), and White Matter (WM). This segmentation is performed using the K-Means clustering algorithm as shown through the following equation:

$$\arg \min_C \sum_{i=0}^2 \sum_{x \in C_i} \|I(x) - \mu_i\|^2, \quad (1)$$

where $C = \{C_0, C_1, C_2\}$ are cluster assignments, $I(x)$ is the normalized intensity at voxel x , and μ_i are cluster centroids.

The underlying principle is that in T1-weighted sMRI, these tissues exhibit distinct intensity profiles. The cluster centroids are then automatically sorted by intensity to assign anatomical labels:

1. **Cluster 0 (Lowest Intensity Centroid)** Corresponds to CSF, which appears darkest in T1 images.
2. **Cluster 1 (Medium Intensity Centroid)** Corresponds to Gray Matter.
3. **Cluster 2 (Highest Intensity Centroid)** Corresponds to White Matter, which appears brightest.

The next phase in the pre-processing pipeline was the volumetric and intensity-based feature extraction. The high-dimensional, segmented 3D MRI volumes were transformed into a low-dimensional, quantitative feature vector for each patient. This helps to learn discriminative patterns. The preprocessing pipeline is illustrated in Figure 2.

3.1.1 Volumetric Features

These features are designed to capture macro-level structural changes and brain atrophy, which are key biomarkers of neurodegeneration. The features computed include absolute volume values like csf volume, gm volume, and wm volume, the total brain volume which is the sum of the above mentioned values and finally the normalized volume ratios such as csf ratio, gm ratio, wm ratio and gm-wm-ratio (gray matter to white matter ratio).

3.1.2 Intensity-based Statistical Features

To capture textural and distributional properties within each tissue type, we extract first-order intensity statistics from the normalized voxel values within each of the three tissue masks:

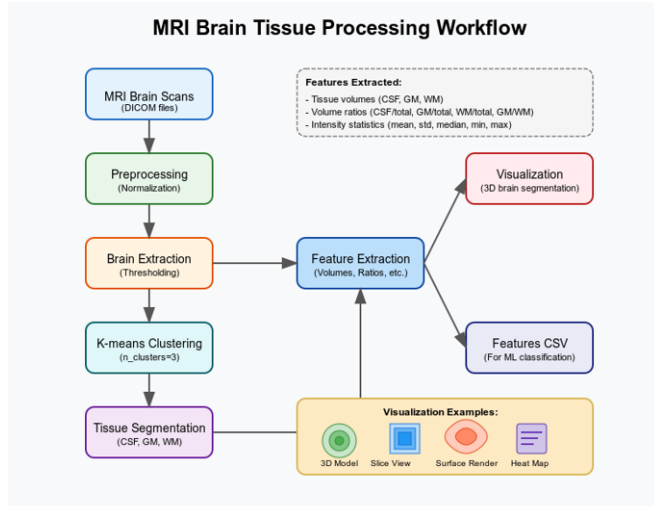


Fig. 2: Brain tissue preprocessing pipeline showing the stages of extraction and segmentation.

- **Measures of Central Tendency:** Mean and median intensity.
- **Measures of Dispersion:** Standard deviation of intensity.
- **Range Indicators:** Minimum and maximum intensity values.

This entire process transforms each 3D sMRI scan into a structured row in a tabular dataset, which is then combined with demographic and clinical data and saved into a csv file. Table 2 summarizes the complete set of extracted features.

Table 2: Summary of Extracted Features from sMRI Scans

Category	Feature Name	Count
Volumetric	CSF Volume, GM Volume, WM Volume	3
	Total Brain Volume	1
	CSF Ratio, GM Ratio, WM Ratio	3
	GM-WM Ratio	1
Intensity (per tissue)	Mean, Median Intensity	6
	Standard Deviation	3
	Min, Max Intensity	6
Total MRI Features		23

Figure 3 shows an example of the segmented brain tissues.

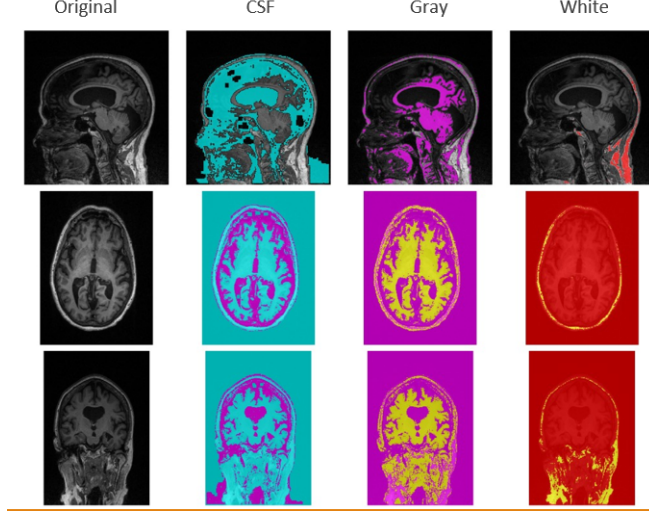


Fig. 3: Visualization of segmented brain tissues: Cerebrospinal Fluid (CSF), Gray Matter (GM), and White Matter (WM).

3.2 Hypergraph Construction

For the task of classifying patients into the six different classes as mentioned above, we employ a Hypergraph Neural Network. Unlike traditional graphs that model pairwise relationships, hypergraphs can capture higher-order relationships among a set of nodes. In our framework, we construct a feature-centric hypergraph. Each of the engineered features (both volumetric and statistical) is treated as a node in the hypergraph. Hyperedges are constructed based on patient similarity which allows the model to learn complex, non-linear interactions between small groups of features simultaneously. This construction yields a hypergraph incidence matrix $\mathbf{H} \in \mathbb{R}^{N \times M}$:

$$\mathbf{H}(i, j) = \begin{cases} 1, & \text{if patient } i \text{ is in hyperedge } j, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The degree of node i is $d_i = \sum_j \mathbf{H}(i, j)$, and the degree of hyperedge j is $\delta_j = \sum_i \mathbf{H}(i, j)$.

This incidence matrix is then converted into an edge-index tensor suitable for the torch-geometric library.

3.3 HyperGNN Model and Training

Our classification model is a deep HyperGNN composed of three consecutive HypergraphConv layers. The hypergraph convolutional layer performs information aggregation through hyperedges. The layer updates node representations by aggregating

information from connected hyperedges.

$$\mathbf{X}^{(l+1)} = \sigma \left(\mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{D}_e^{-1} \mathbf{H}^T \mathbf{D}_v^{-1/2} \mathbf{X}^{(l)} \mathbf{W}^{(l)} \right), \quad (3)$$

where $\mathbf{X}^{(l)} \in \mathbb{R}^{N \times d_l}$ is the node feature matrix at layer l , $\mathbf{D}_v \in \mathbb{R}^{N \times N}$ is the diagonal node degree matrix, $\mathbf{D}_e \in \mathbb{R}^{M \times M}$ is the diagonal hyperedge degree matrix, $\mathbf{W}^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$ is the learnable weight matrix, and σ is the ReLU activation function.

Each HypergraphConv layer is followed by a BatchNorm1d layer for normalization, a ReLU activation function to introduce non-linearity, and a Dropout layer to prevent overfitting. Residual Connections are incorporated between the convolutional blocks. The output of each block is added to its input, which helps mitigate the vanishing gradient problem in deep architectures and promotes more stable training. The final layer is a Multi-Layer Perceptron (MLP) classifier head that maps the learned high-level feature representations to the final class predictions. The complete architecture is detailed in Table 3.

Table 3: HyperGNN Model Architecture

Layer	Type	Output Dim	Parameters
1	HypergraphConv	128	Input $\rightarrow$ 128
	BatchNorm1d + ReLU + Dropout	128	$p = 0.3$
2	HypergraphConv	64	128 $\rightarrow$ 64
	BatchNorm1d + ReLU + Dropout	64	$p = 0.3$
	Residual Connection	64	—
3	HypergraphConv	32	64 $\rightarrow$ 32
	BatchNorm1d + ReLU + Dropout	32	$p = 0.3$
	Residual Connection	32	—
4	MLP Classifier	6	32 $\rightarrow$ 6

The model is trained using the Adam optimizer with a learning rate of 0.01. For the loss function we use a weighted Cross-Entropy Loss function as shown in (4). This helps address the class imbalance.

$$\mathcal{L} = - \sum_{i=1}^N \sum_{c=1}^C w_c \cdot y_{i,c} \log(\hat{y}_{i,c}), \quad (4)$$

where $w_c = N/(C \cdot N_c)$ is the weight for class c , N_c is the number of samples in class c , and $\hat{y}_{i,c}$ is the predicted probability for sample i and class c .

The model is trained for 500 epochs with a batch size of 32, using an 80%/20% train/test split. For evaluation, in addition to standard accuracy metrics, we assess the model’s robustness by introducing a small amount of Gaussian noise to the test set features before prediction.

3.4 Disease Severity Modelling

Beyond multi-class classification of the disease class, we implement a secondary parallel severity assessment pipeline using clustering-based approaches.

1. Severity Clustering: Given the continuous nature of neurodegeneration, we apply k-means clustering ($k=3$) on standardized MRI and clinical features. This step groups patients into three distinct clusters based on similarities in their overall feature profiles.

2. Severity Level Mapping: The assumption is that the clusters obtained correspond to different levels of disease severity. To assign meaningful clinical labels to these clusters, we calculate the L2-norm (magnitude) of each cluster’s centroid. The centroids are then ordered by their norm, and labels are assigned accordingly as mild, moderate and severe.

3. Severity Prediction: After generating these severity labels in an unsupervised manner, we train a Random Forest Classifier to predict the severity cluster for any given patient based on their input features.

3.5 Explainability and Interpretability

To ensure our framework is not a "black box" and to provide insights and back the decisions of the model to improve confidence for clinicians, we integrated SHAP (SHapley Additive exPlanations) for model explainability.

1. Explaining the HyperGNN Classification Model: For the HyperGNN model, we utilize shap.DeepExplainer. This method calculates SHAP values to quantify the contribution of each input feature (e.g., GM volume, CSF mean intensity) towards the model’s output for each diagnostic class. The resulting SHAP summary plots provide a global view of feature importance, highlighting the key biomarkers that the model learns to rely on for its predictions.

2. Explaining the Random Forest Severity Model: For the Random Forest model which is used for severity analysis, we use the optimized shap.TreeExplainer. This allows us to determine which features are most influential in separating the severity groups. This helps to validate the finding of the model.

4 Results

In this section we present the results obtained from applying our proposed Hypergraph Neural Network (HyperGNN) framework for neurodegenerative disease classification and the parallel machine learning approach for severity modeling. The performance metrics demonstrate the effectiveness of our methodology on the processed ADNI dataset features.

The HyperGNN model was trained for 200 epochs to classify subjects into six different diagnostic categories based on the engineered volumetric and intensity features derived from sMRI and demographic data. The training process exhibited steady convergence, with decreasing loss and increasing training accuracy over epochs. The initial Cross-Entropy Loss at the first training epoch was 1.8112, with a training accuracy of 12.23%. The loss decreased throughout the training process and was about 0.09 at the

end of the training. Training accuracy also continues to improve over the training process from about 12% in the first epoch to around 90% at the half way stage to ending at about 97% at the end of the process. Figure 4 illustrates the training progression.

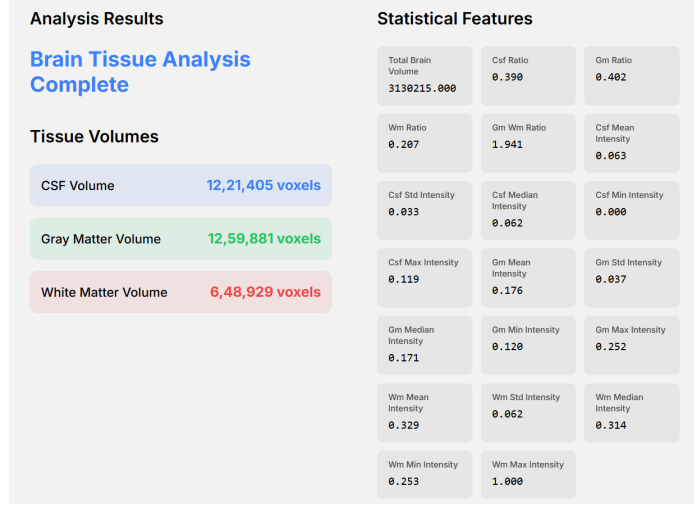


Fig. 4: Training progression of the HyperGNN model showing loss convergence and accuracy improvement.

To assess the model’s robustness and generalization capability, the trained HyperGNN was evaluated on a held-out test set with added Gaussian noise applied to the input features. The model achieved a weighted average precision, recall, and F1-score of 0.94 across all six diagnostic categories, demonstrating its ability to maintain high performance even with slight perturbations in the input data and generalize well on unseen data. Table 4 presents the detailed classification performance.

Table 4: HyperGNN Classification Performance by Diagnostic Category

Class	Precision	Recall	F1-Score	Support
CN	0.94	0.96	0.95	892
SMC	0.92	0.91	0.91	523
MCI	0.93	0.92	0.92	687
EMCI	0.94	0.93	0.93	745
LMCI	0.95	0.94	0.94	612
AD	0.96	0.97	0.96	879
Macro Avg	0.94	0.94	0.94	4338
Weighted Avg	0.94	0.94	0.94	4338

Beyond this classification process, we also implemented the severity assessment through clustering-based approaches. K-Means clustering algorithm was used on the scaled feature set to identify distinct patient subgroups as mild, moderate and severe, potentially corresponding to disease severity levels. A Random Forest classifier was subsequently trained to predict these data-driven severity clusters using the input features. The classifier demonstrated perfect performance on the training data, achieving almost perfect accuracy, precision, recall, and F1-score across all three severity clusters. This suggests that the feature set contains sufficient information to reliably distinguish between the identified severity groups. The detailed severity classification results are presented in Table 5. Figure 5 shows the web application interface displaying prediction results with class probabilities and severity assessment.

Table 5: Severity Classification Results

Severity Class	Precision	Recall	F1-Score	Support
Mild	0.99	0.98	0.99	1716
Moderate	0.98	0.99	0.99	1557
Severe	0.99	0.99	0.99	2065
Accuracy	—	—	0.99	5338
Macro Avg	0.99	0.99	0.99	5338
Weighted Avg	0.99	0.99	0.99	5338

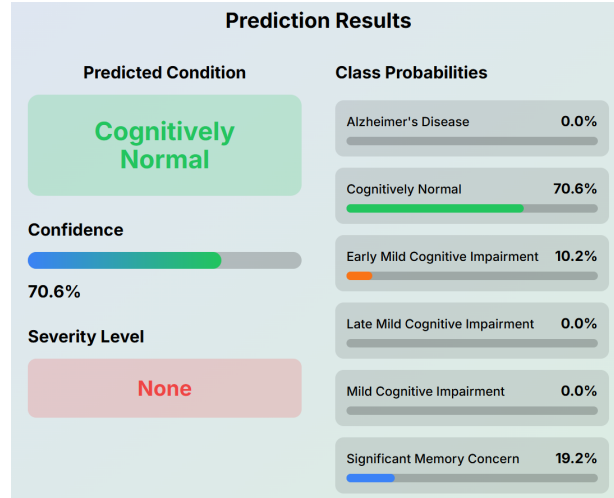


Fig. 5: Web application interface showing prediction results with class probabilities, confidence score, and severity level.

To ensure transparency and clinical interpretability, SHAP analysis was performed on both the classification and severity models. For the classification process, SHAP summary plots were generated for each class, providing insights into which volumetric and intensity features were most influential in the HyperGNN’s classification decisions. For the severity classifier as well, the SHAP summary plots were generated highlighting the key features driving the distinction between the Mild, Moderate, and Severe patient clusters. These explainability results provide valuable information regarding the biomarkers prioritized by the models, enhancing their potential for clinical trust and application. Figure 6 shows SHAP summary plot for the AD class.

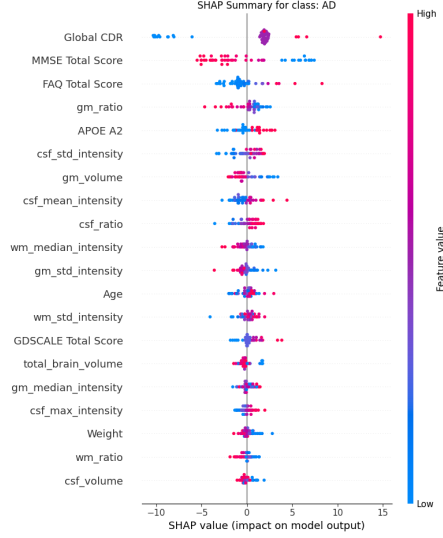


Fig. 6: SHAP summary plot showing feature importance for Alzheimer’s Disease classification.

5 Conclusion

In this study, we built a multi-modal hypergraph neural network framework, for the multi-class classification of the neurodegenerative disorders, with specific application to Alzheimer’s Disease diagnosis using the ADNI dataset. We also worked on the severity analysis along with a focus on the explainability to provide trustable results. We obtained a very robust overall performance and results with 94% weighted average F1-score even after adding noise perturbation while testing, showing reliable generalization.

The HyperGNN framework offers several clinical advantages including accessibility as it requires only structural MRI and routine clinical data. It also supports early

detection with the help of severity analysis, helping in early intervention. Explainable predictions build clinician trust and enable verification of clinical reasoning.

Although our approach does not provide state of the art results, but our integration of explainability, unlike most of the related work, proves as an advantage in addressing clinical requirements for interpretable decision-making. Also the usage of HyperGraphs helps uniquely model inter-patient relationships, capturing population-level patterns unavailable to comparison methods. This along with combining the tabular patient data with the MRI dataset, could provide as a base for personalization in terms of diagnosis.

Future work could focus on validating this framework on larger, independent datasets. Additional data modalities could also be incorporated. Study on finding optimal hypergraph construction strategies could also yield performance improvements. Overall, this study underscores the potential of advanced graph-based deep learning and explainable AI in tackling the complex challenges in neurodegenerative disease diagnostics.

References

- Adarsh, V., Gangadharan, G.R., Fiore, U., *et al.*: Multimodal classification of alzheimer’s disease and mild cognitive impairment using custom mkscdl kernel over cnn with transparent decision-making for explainable diagnosis. *Scientific Reports* **14**, 1774 (2024) <https://doi.org/10.1038/s41598-024-52185-2>
- El-Assy, A.M., Amer, H.M., Ibrahim, H.M., *et al.*: A novel cnn architecture for accurate early detection and classification of alzheimer’s disease using mri data. *Scientific Reports* **14**, 3463 (2024) <https://doi.org/10.1038/s41598-024-53733-6>
- Ghorbani, M., Kazi, A., Soleymani Baghshah, M., Rabiee, H.R., Navab, N.: Ra-gcn: Graph convolutional network for disease prediction problems with imbalanced data. *Medical Image Analysis* **75**, 102272 (2022) <https://doi.org/10.1016/j.media.2021.102272>
- Hao, X., Li, J., Ma, M., *et al.*: Hypergraph convolutional network for longitudinal data analysis in alzheimer’s disease. *Computers in Biology and Medicine* **168**, 107765 (2024) <https://doi.org/10.1016/j.compbimed.2023.107765>
- Kim, M., *et al.*: Interpretable temporal graph neural network for prognostic prediction of alzheimer’s disease using longitudinal neuroimaging data. In: 2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Houston, TX, USA, pp. 1381–1384 (2021). <https://doi.org/10.1109/BIBM52615.2021.9669504>
- Lao, H., Jia, H., Chen, Z.: Diagnosis of alzheimer’s disease based on structural graph convolutional neural networks. In: Proceedings of the ACM Turing Award Celebration Conference - China 2024 (ACM-TURC ’24), pp. 148–152 (2024). <https://doi.org/10.1145/3674399.3674453>

- Ma, D., Lu, D., Popuri, K., Wang, L., Beg, M.F.: Differential diagnosis of frontotemporal dementia, alzheimer’s disease, and normal aging using a multi-scale multi-type feature generative adversarial deep neural network on structural magnetic resonance images. *Frontiers in Neuroscience* **14**, 853 (2020) <https://doi.org/10.3389/fnins.2020.00853>
- Pan, J., Lei, B., Shen, Y., Liu, Y., Feng, Z., Wang, S.: Characterization multimodal connectivity of brain network by hypergraph gan for alzheimer’s disease analysis. In: *Pattern Recognition and Computer Vision: 4th Chinese Conference, PRCV 2021*, pp. 467–478. Springer, ??? (2021)
- Subaramya, S., Kokul, T., Nagulan, R., Pinidiyaarachchi, U.A.J.: Graph neural network based alzheimer’s disease classification using structural brain network. In: *2022 22nd International Conference on Advances in ICT for Emerging Regions (ICTer)*, Colombo, Sri Lanka, pp. 1–6 (2022). <https://doi.org/10.1109/ICTer58063.2022.10024076>
- Sang, Y., Li, W.: Classification study of alzheimer’s disease based on self-attention mechanism and dti imaging using gcn. *IEEE Access* **12**, 24387–24395 (2024) <https://doi.org/10.1109/ACCESS.2024.3364545>
- Tang, Y., Xiong, X., Tong, G., *et al.*: Multimodal diagnosis model of alzheimer’s disease based on improved transformer. *BioMedical Engineering OnLine* **23**, 8 (2024) <https://doi.org/10.1186/s12938-024-01204-4>